

Language-driven Diversified Image Retargeting

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ABSTRACT

Content-aware image resizing could automatically retarget an image to different aspect ratios while preserving visually salient contents. However, it is difficult for users to interact with the retargeting process and control the results. In this paper, we propose a language-driven diversified image retargeting (LDIR) method that allows the users to control the retargeting process by providing additional textual descriptions. Taking the original image and user-provided texts as inputs, LDIR retargets the image into the desired resolution while preserving the content indicated by texts. Following a self-play reinforcement learning pipeline, a multimodal reward function is proposed by considering both the visual quality and language guidance. Preliminary experiments manifest that LDIR can achieve diversified image retargeting guided by texts.

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1 INTRODUCTION

Existing retargeting approaches focus on improving the output quality in terms of visual appearance and similarity preservation with the input image but neglect the personal control of the results. Manually customized retargeting images is time-consuming and difficult for non-professional users. In this paper, we design a diverse image retargeting pipeline which supports additional texts to guide the retargeting process. Different from Liu et al. [2018] which uses semantic components as supplements of the inputs, we propose to utilize texts as descriptions for the output images. By using different textual descriptions, we can acquire significantly different image contents where semantics are equal to the user-providing descriptions. However, it is challenging to involve the text information in the retargeting processes which are mainly driven by visual importance and similarity calculation.

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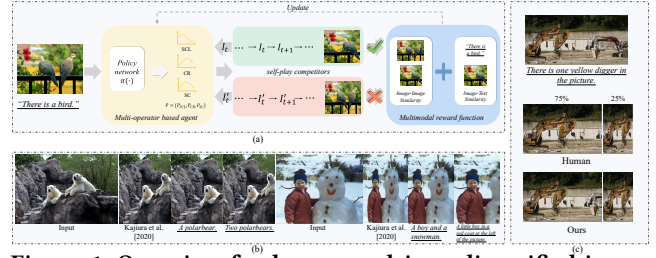


Figure 1: Overview for language-driven diversified image retargeting. (a) the proposed reinforcement learning pipeline, which involves a multi-operator-based agent and a multimodal reward function; (b) comparisons with [Kajiura et al. 2020] when reducing 50% width; (c) results on more target ratios and comparisons with human-made results by professional software.

Recently, the issue of language-driven image editing/generation has been brought to public attention. The success of the vision-language pre-training models (e.g., CLIP [Radford et al. 2021]) has built a bridge for text and image semantic multimodal alignment. In this paper, we propose a simple yet efficient language-driven diversified image retargeting (LDIR) method based on multimodal pretraining models to measure the semantic difference between input texts and retargeted results. We employ a self-play reinforcement learning pipeline and obtain a CLIP-based multimodal reward function to optimize the agent in reinforcement learning. Experimental results manifest that LDIR can perform effective and extensive image retargeting based on different text conditions.

2 METHODS

As shown in Figure 1 (a), we propose a reinforcement learning pipeline based on multi-operator image retargeting methods [Song et al. 2018] to control the resizing process by user-provided texts. A *multi-operator based agent* and a *multimodal reward function* are adopted to generate the optimal sequence of operators. The multi-operator agent works as a decision maker and interacts with the reward function to select the operator step by step. The multimodal reward works as an assessment to evaluate the selected operator sequence and update the agent network.

Multi-operator based Agent. Given an input image I of size (w, h) , the target size (w', h') , and conditioned texts C_t , the agent determines the multi-operator sequence to retarget I by multiple-step predictions. Without loss of generality, we discuss the case when only the width of I is reduced from w to w' . Then, the length of the multi-operator sequences is $t_{max} = w - w'$. Three kinds of retargeting operators are involved in forming the action space: cropping (CR), scaling (SCL), and seam carving (SC). Based on a policy network $\pi(\cdot)$, the agent calculates the possibilities of adopting different operators for a given input image. For step $t + 1$, the agent receives the retargeted image I_t at step t and then outputs the probability distribution of three operators to sample an operator for further retargeting I_t :

$$\mathbf{P}_t = \{p_{cr}, p_{scl}, p_{sc}\}_t = \pi(I_t; \theta_\pi), \quad (1)$$

where $\mathbf{P}_t$ denotes the probability distribution of three operators at step $t + 1$. Then we get the output for step $t + 1$ by $I_{t+1} = F(I_t, \mathbf{P}_t)$. The policy network $\pi(\cdot)$ is updated by the following goal:

$$L_t^\pi = -r_{mul}(I_t, C_t) - H(\pi(I_t; \theta_\pi)), \quad (2)$$

where $r_{mul}(\cdot)$ measures the similarity between the generated image I_t with the input image and the text condition. $H(\cdot)$ is a function that calculates entropy, which prevents agent to a local optimum. The agent works t_{max} times until the output ratio satisfies the target ratio.

Multimodal Reward Function. The multimodal reward function $r_{mul}(\cdot)$ we adopted is defined as following:

$$r_{mul}(I, I', C_t) = -BDW(I, I') + \beta \times \cos(CLIP_{image}(I'), CLIP_{text}(C_t)), \quad (3)$$

where $BDW(\cdot)$ is the image similarity measurement following Rubinstein et al. [2009], which is used to evaluate the distance between the retargeted image and the original image. The second item in Eq. (3) measures the similarity between generated images and the given texts by multimodal CLIP encoders. $CLIP_{image}(\cdot)$ and $CLIP_{text}(\cdot)$ represent the CLIP image and text encoders. $\cos(\cdot)$ measures the cosine similarity between multimodal embeddings.

To guarantee the selected operator sequence is optimal, we introduce the self-play procedure. The agent predicts two operator sequences simultaneously and obtains two results in a self-play way. We judge the final retargeted results by $r_{mul}(\cdot)$. When training, we return the victor's reward to the update agent by Eq. (2). For testing, we use the victor as the final output.

3 RESULTS AND DISCUSSION

Implementation. We randomly select 3,000 image-description pairs from Microsoft COCO as training data. By calculating the similarity of image and description, the model learns to understand the semantics of texts and images. The batch size is set to 16. During training, the step number t_{max} is sampled randomly from $\{1, \dots, 20\}$ for each training image. As for inference, we determine $t_{max} = \frac{w-w'}{10}$ for accelerating, thus the operation predicted for each step will be conduct ten times.

Results. We select some images from COCO and use all images in RetargetMe as the test dataset. We set the target ratio to 50% of the original size and compare the proposed LDIR with Kajiura et al.

Table 1: Voting statistics for the user study.

	Ours	Kajiura et al. [2020]	Both
Visual quality	62%	14%	24%
Text consistency	74%	2%	24%

[2020]. Figures 1(b) and 2 show the visual comparisons. A user study is conducted for assessment. We display twenty groups of results, including the original image, text, and retargeted results of Kajiura et al. [2020] and LDIR. Participants are asked to compare the visual quality and select the retargeted images which correspond to the given texts. Table 1 shows the results. Overall, our method could not only generate images consistent with text guidance but also improve the visual quality. The reason is that the text conditions are frequently connected to image contents and could serve as supplementary information to assist in understanding the image contents. Furthermore, we also extend visual experiments on target ratios of 25% and 75%, as shown in Figure 1(c).

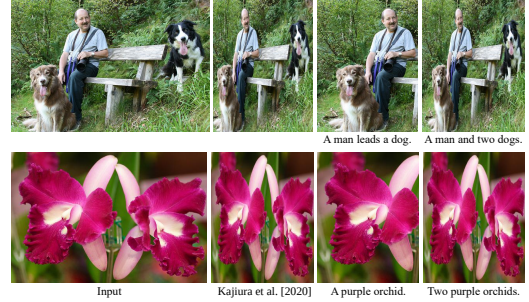


Figure 2: Visual comparisons when the target ratio is 50%.

Conclusion and Discussion. This study proposes a reinforcement learning framework for diversified image retargeting guided by text. The experimental findings reveal that LDIR achieved stable and accurate results guided by texts. When multiple similar objects exist in the original images (the 2nd row in Figure 2), it is hard to control which object to preserve in the target image. In the future, we will consider modeling the spatial relation in the source/target images and specific such relation with complex text conditions.

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